Week #8 : Ecological Systems - Population Change, Growth & Reproductive Strategies

Subject/الموضوع: Pre-AP Biology Unit Title: Ecological Systems Grade9: الصف

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1	I can explain how births, deaths, immigration, and emigration change population size. I can calculate net population change and distinguish growth, decline, and stability.	1.1 Flexible display of information (size, color, font, layout).	Lesson 11A PowerPoint: Population Change — Four processes, net-change formula and population data table. Review Teams / Grade 9 Pre-AP Biology / General channel
Lesson 2 الحصة الثانية	I can interpret population-change data to evaluate recovery or risk. I can justify a conservation recommendation using births, deaths, immigration, and emigration evidence.	5.3 Build fluency with graduated scaffolds and models.	Lesson 11A PowerPoint: Population Change — UAE oryx wildlife-reserve decision, evidence-based conservation and AP-style questions. Review Teams / Grade 9 Pre-AP Biology / General channel
Lesson 3 الحصة الثالثة	I can use population data and graphs to describe changing growth rates. I can distinguish an observed growth pattern from possible biological explanations.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Lesson 11B PowerPoint: Population Growth Patterns — J-shaped growth, monthly growth data, UAE oryx trends and peer review. Review Teams / Grade 9 Pre-AP Biology / General channel
Lesson 4 الحصة الرابعة	I can explain how resources and density-dependent or density-independent limiting factors affect population growth. I can use data to explain carrying capacity and population fluctuations.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Lesson 12 PowerPoint: Limiting Factors & Carrying Capacity — gazelle investigation, rainfall data, exponential and logistic growth. Review Teams / Grade 9 Pre-AP Biology / General channel
Lesson 5 الحصة الخامسة	I can compare r-selected and K-selected reproductive strategies. I can use life-history evidence to predict which strategy is favored under different environmental conditions.	8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Lesson 13 PowerPoint: Reproductive Strategies — r–K continuum, species evidence, UAE desert mammal and annual plant cases. Review Teams / Grade 9 Pre-AP Biology / General channel

Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	• Biology notebook and calculator required for population-change calculations, graphs, evidence-based explanations and comparison tasks. • Review PowerPoint resources on Microsoft Teams. • Reinforce the distinction between observations, calculations and justified predictions.

الأسبوع# Week #8

Technology Integration دمج التكنولوجيا	• PowerPoint data tables, population-growth graphs, carrying-capacity models and life-history evidence. • Microsoft Teams for lesson resources and review; optional digital graphing.
Cross-curricular links الروابط بين المواد الدراسية	• Mathematics: net change, differences between time intervals, graph interpretation and numerical evidence. • Geography/Environmental Science: UAE reserves, rainfall, drought, resource limits and species conservation.
External Assessment Integration دمج التقييمات الخارجية	Pre-AP Biology / College Board Unit 1: Ecological Systems — ECO 2 Population Dynamics. AP-style calculations, data interpretation, modeling and evidence-based justifications.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Thinking: quantitative reasoning, cause-effect, comparing patterns and drawing inferences. • Communication: CER explanations, scientific vocabulary and peer feedback. • Collaboration: data investigations and reproductive-strategy comparisons.
Standards المعايير	• Lessons 1–3: ECO 2.2(a), ECO 2.2(b); EK 2.2.1a, 2.2.1d. • Lesson 4: ECO 2.2(a–c); EK 2.2.1a–d. • Lesson 5: ECO 2.2.2(a–b): reproductive strategy and population growth.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Lesson 1: How can a population increase, decrease or remain stable when individuals enter and leave? Lesson 2: What evidence can help reserve managers decide whether a population is recovering? Lesson 3: What can the shape of a population graph tell us about the rate of change? Lesson 4: Why can a population grow quickly for a while but not forever? Lesson 5: Why can both many offspring and greater care for fewer offspring be successful?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 — Population Change Calculations MUST: Identify births, deaths, immigration and emigration. SHOULD: Calculate net change and ending population. COULD: Explain growth, decline or stability using data. Lesson 2 — UAE Reserve Decisions MUST: Identify additions and removals in reserve data. SHOULD: Calculate and compare net change in two years. COULD: Justify a conservation action with numerical evidence. Lesson 3 — Growth Patterns MUST: Identify a rapidly increasing population pattern. SHOULD: Calculate interval increases and interpret a graph. COULD: Distinguish evidence from possible causes of changing growth rates.

	Lesson 4 — Limiting Factors and Carrying Capacity MUST: Define limiting factors and carrying capacity. SHOULD: Distinguish density-dependent and independent factors. COULD: Use rainfall and population data to explain changing carrying capacity. Lesson 5 — Reproductive Strategies MUST: Identify r- and K-associated traits. SHOULD: Compare offspring number, maturity, lifespan and care. COULD: Predict which strategy may be favored after disturbance and justify with evidence.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام Heritage التراث
	المجال Area: Culture / Sustainability / Conservation
	• UAE oryx reserves: analyze births, deaths and movement to evaluate recovery and conservation risks. • UAE wildlife monitoring: interpret population trends without assuming causes from graphs alone. • UAE desert habitats: connect rainfall, food availability and drought to gazelle carrying capacity. • UAE biodiversity: compare life-history strategies of desert mammals and annual plants under variable rainfall.